



REPORT
ON
DELHI WEATHER

SUBMITTED TO:

Mr. Rachit Chauhan

SUBMITTED BY:

Mahek

(GU-2025-8852)

MBA Business Analytics (Batch 2025-2027)

GNA UNIVERSITY, PHAGWARA, PUNJAB

Contents

INTRODUCTION	3
Problem Statement	3
Objectives.....	3
Integrated Development Environment (IDE)	3
Google Colab	3
Libraries Used.....	4
ABOUT THE DATASET.....	4
METHODOLOGY	7
Data Preprocessing	7
Exploratory Data Analysis (EDA)	8
Statistical Analysis	8
Location-Based Analysis.....	8
Visualization Techniques	8
IQR (Interquartile Range) Outlier Detection Method	9
DATA VISUALIZATION.....	10
Analysis of each column.....	10
Correlation	17
Monthly analysis	18
Location based analysis.....	20
RECOMMENDATIONS	21
CONCLUSION	22

INTRODUCTION

Air pollution has become a major environmental and public health concern, especially in urban areas. The Air Quality Index (AQI) is a standardized indicator used to measure and communicate air quality levels. Weather parameters such as temperature, humidity, wind speed, and pressure significantly influence air pollutant dispersion and concentration.

Problem Statement

Rapid urbanization and industrialization have led to deteriorating air quality. Understanding how weather conditions affect AQI is essential for pollution control and policy planning.

Objectives

- To analyze weather patterns in Delhi.
- To evaluate air quality levels by analyzing AQI values and major pollutants.
- To identify relationships between weather parameters and air pollution.
- To perform exploratory data analysis (EDA) to uncover trends and patterns.
- To visualize temporal trends in weather and air quality data.
- To draw meaningful insights and conclusions that can help to understand environmental conditions and pollution behavior in Delhi.

Integrated Development Environment (IDE)

Google Colab

Google Colaboratory (Google Colab) is a cloud-based IDE that allows users to write and execute Python code directly in a web browser without requiring local installation.

Advantages of Google Colab

- No setup or configuration required
- Free access to computational resources
- Easy integration with Google Drive for data storage
- Supports Jupyter Notebook (.ipynb) format
- Enables quick sharing and collaboration

Programming Language

Python is used for data analysis and visualization due to its simplicity and extensive ecosystem of scientific libraries.

Libraries Used

NumPy: Used for numerical operations and array-based computations.

Pandas: Used for data preprocessing, handling missing values, data aggregation, and statistical analysis.

Matplotlib: Used for creating line plots, bar charts, and other basic visualizations.

Seaborn: Used for advanced statistical visualizations such as box plots, violin plots, and heatmaps.

Role in the Analysis

Google Colab provided an efficient and accessible platform for:

- Data cleaning and preprocessing
- Exploratory data analysis (EDA)
- Visualization of weather and air quality patterns
- Studying relationships between meteorological parameters and pollution indicators

ABOUT THE DATASET

This dataset contains hourly weather and air pollution data for Delhi, collected for different monitoring locations. It combines meteorological parameters with air quality indicators, making it suitable for environmental analysis, pollution studies, and weather–AQI relationship analysis.

Dataset Overview

- Total records: 52,566
- Total variables: 16
- Time frequency: Hourly
- Data type: Time-series + spatial + environmental data

Temporal Information: These columns allow daily, monthly, seasonal, and time-series analysis.

Column	Description
date_ist	Date of observation (Indian Standard Time)
time_ist	Hour of observation (Indian Standard Time)

Location Information: Useful for spatial analysis, heatmaps, and location-based comparisons.

Column	Description
location	Monitoring station name
lat	Latitude
lon	Longitude

Weather Parameters: These variables help analyze how weather affects pollution levels.

Column	Description
temp_c	It indicates how hot or cold the atmosphere is.
humidity	Tells how much moisture (water vapor) is present in the air.
pressure_mb	It shows how heavy or light the air above a place is.
windspeed_kph	Tells how fast the air is moving from one place to another.
condition_text	Tells the overall state of the atmosphere at a particular place and time.
Description WMO	Scientific and globally accepted explanation of weather conditions.

Pressure(mb):

Below 1000 mb → Very low pressure
(Unstable weather, storms, strong winds)

1000–1010 mb → Low pressure
(Cloudy conditions, chances of rain)

1010–1020 mb → Normal pressure
(Stable, typical weather)

1020–1030 mb → High pressure
(Clear skies, dry and calm conditions)

Above 1030 mb → Very high pressure
(Very stable air, heat trapping, poor pollutant dispersion)

Humidity (%):

Below 30% → very dry
(skin feels dry, dust stays in air)

30–40% → Air is a little dry

40–60% → comfortable(best range)

60–70% → slightly sticky

70–80% → very sticky (fog)

Above 80% → too humid / wet
(heavy air, uncomfortable)

temp c (Temperature in °C):

0 – 10°C → Very cold
10 – 20°C → Cool
20 – 25°C → Pleasant/Mild
25 – 30°C → Warm
30 – 35°C → Hot
35 – 40°C → Very hot
Above 40°C → Extreme heat/ Heatwave

Windspeed(km/hr):

Below 2 km/hr → Calm air
(Stagnant conditions, pollutants accumulate)
2–5 km/hr → Very light wind
(Limited dispersion of pollutants)
5–10 km/hr → Light wind
(Some air movement, partial pollutant dispersion)
10–20 km/hr → Moderate wind
(Good ventilation, pollutants disperse effectively)
20–30 km/hr → Strong wind
(Rapid dispersion, dust resuspension possible)
Above 30 km/hr → Very strong wind
(Excellent dispersion, dust storms / turbulence possible)

AQI:

Good: 0 – 50	Very Poor: 201 – 300
Moderate: 51 – 100	Severe / Hazardous: 301 – 500
Poor: 101 – 200	

Air Quality & Pollution Indicators: These are key indicators of air pollution and health risk.

Column	Description
aqi_index	tells how clean or polluted the air is
pm2_5	tells us how much fine particulate pollution is present in the air
pm10	indicates the concentration of inhalable coarse particulate matter in the air
Co	indicates the intensity of incomplete combustion-related pollution
no2	reflects pollution from combustion sources and indicates potential respiratory health risks

METHODOLOGY

Data Preprocessing

Data preprocessing was performed to ensure accuracy and reliability:

- Duplicate records were identified and removed.
- Missing values were handled using appropriate statistical methods.
- Date and time fields were formatted for temporal analysis.
- Outliers were detected using the Interquartile Range (IQR) method and evaluated to retain meaningful extreme pollution events.
- **Duplicate Rows or columns:**
- There are 6 duplicate rows present in the dataset i.e., 6 rows in the dataset are exact copies of earlier rows.
- **Columns with Missing Values**

1. lon	Non-null: 47,310 Missing: 5,256 values (~10%)
2. temp_c	Non-null: 47,310 Missing: 5,256 values (~10%)
3. humidity	Non-null: 47,310 Missing: 5,256 values (~10%)

Error Handling

Duplicate rows : Duplicate rows can significantly bias statistical results, distort data distributions, and mislead trend analysis.

`drop_duplicates()` is used to remove duplicate rows.

Columns with Missing Values

Lon: Missing longitude values were imputed using the location-specific constant longitude, assuming fixed geographic coordinates for each monitoring station.

Temperature: Missing values in temperature were filled using median because of presence of outliers.

Humidity: Missing values in humidity were filled using mean because of absence of outliers.

Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to understand data distribution and variability:

- Descriptive statistics were computed for all variables.
- Box plots and violin plots were used to examine data spread and skewness.
- Monthly and seasonal trends were analyzed using aggregated line plots.

Statistical Analysis

Correlation analysis was performed to examine relationships between meteorological variables and air pollutants. Pearson correlation coefficients were used to identify the strength and direction of associations between weather parameters and AQI and pollutant concentrations.

Location-Based Analysis

A location-wise comparison was carried out across monitoring stations including Anand Vihar, Connaught Place, Dwarka, Okhla Phase III, Rohini, and IGI Airport. Variations in air quality were analyzed in relation to local weather conditions and potential emission sources.

Visualization Techniques

Data visualization was performed using Python libraries to effectively communicate findings:

- Line charts for monthly trends
- Box plots for variability and outlier detection
- Heatmaps for correlation analysis
- Comparative plots for station-wise assessment

IQR (Interquartile Range) Outlier Detection Method

The IQR method is a statistical technique used to identify outliers by focusing on the middle 50% of the data. It works well even when the data are skewed.

Key terms

- Q1 (First Quartile): 25th percentile
- Q3 (Third Quartile): 75th percentile
- IQR: $IQR = Q3 - Q1$

Outlier limits

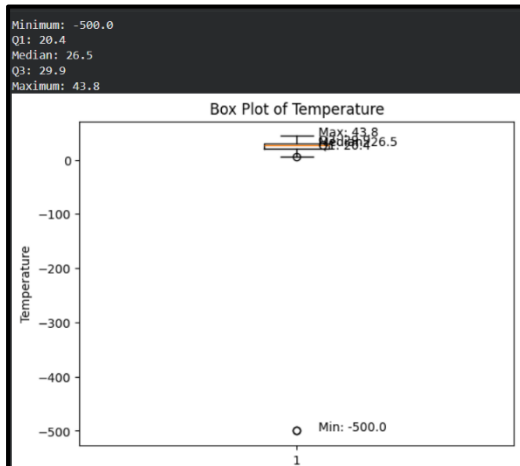
- Lower bound: $Q1 - 1.5 \times IQR$
- Upper bound: $Q3 + 1.5 \times IQR$

Any observation below the lower bound or above the upper bound is treated as an outlier.

DATA VISUALIZATION

Analysis of each column

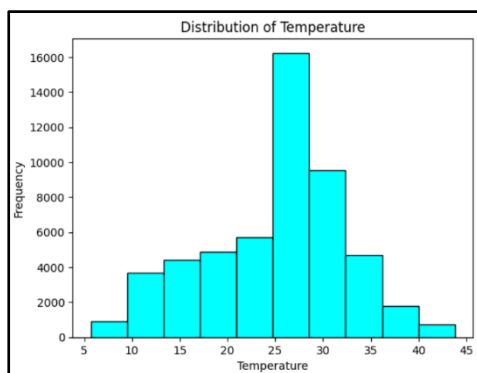
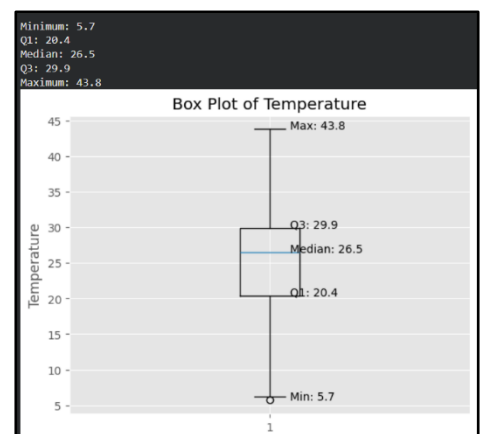
Temperature



(43.8°C), showing extreme summer conditions.

A strong negative outlier at -500°C is observed, which is physically impossible and clearly indicates a data entry or missing-value error.

- The minimum temperature observed is 5.7°C , representing cooler climatic conditions.
- 50% data lies between 20.4 and 29.9, indicating mild to warm weather.
- The maximum temperature



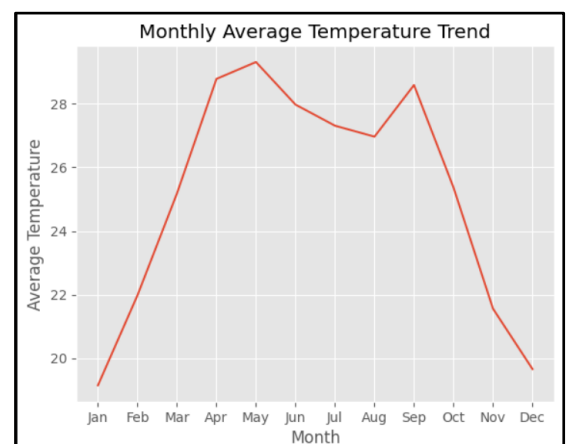
The region experiences mostly moderate to warm weather, with temperatures largely lies between 25 -29($^{\circ}\text{C}$).

The distribution in positively skewed as the right tail is longer (higher temperature values).

1 . Seasonal Pattern

- The graph shows a clear seasonal variation in temperature.
- Temperatures increase steadily from January to May, indicating the transition from winter to summer.
- The highest average temperature occurs around May, representing peak summer conditions.

2. Summer to Monsoon Transition



- After May, temperatures show a gradual decline from June to August.
- This decrease corresponds to the onset of the monsoon season, where cloud cover and rainfall reduce surface heating.
- A slight rise in September suggests post-monsoon warming.

3. Winter Onset

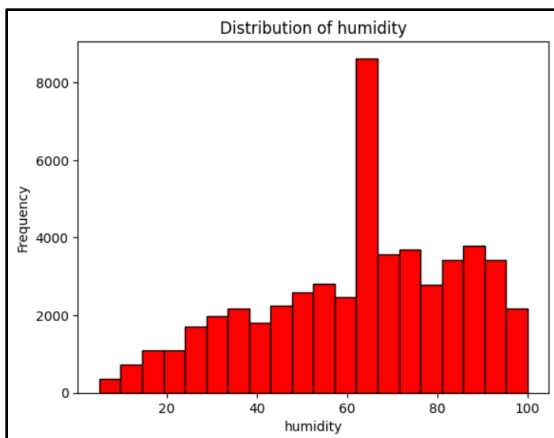
- From October onwards, the temperature drops sharply.
- November and December show the lowest average temperatures, indicating winter conditions.
- December records the minimum monthly average temperature in the dataset.

4. Data Quality Note

- An extreme minimum value of -500°C was observed in the dataset, indicating an outlier or data entry error.
- It were excluded from analysis to ensure accurate trend interpretation.

Humidity

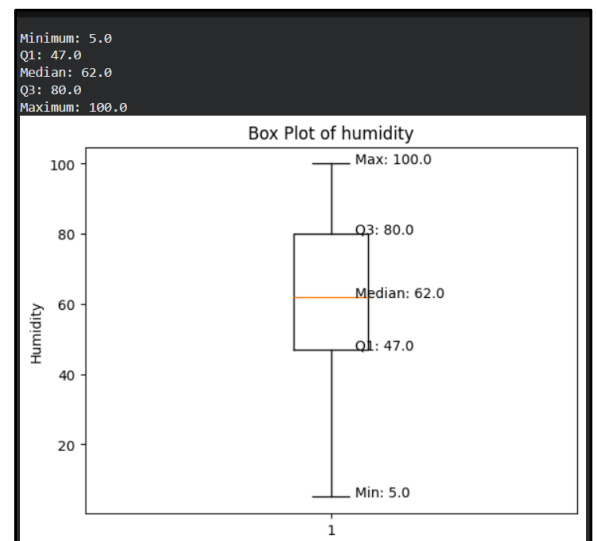
- The minimum humidity recorded is 5%, representing very dry conditions.
- The maximum humidity reaches 100%, indicating fully saturated air, often associated with rainfall or foggy conditions.
- 50% of the humidity values lies between 47-80(%), indicating noticeable variability in atmospheric moisture.



The highest frequency occurs around 60–65% humidity.

This indicates that moderately high humidity is the most common atmospheric condition in the dataset.

The distribution is negatively skewed, Occasional very dry conditions pull the distribution to the left.

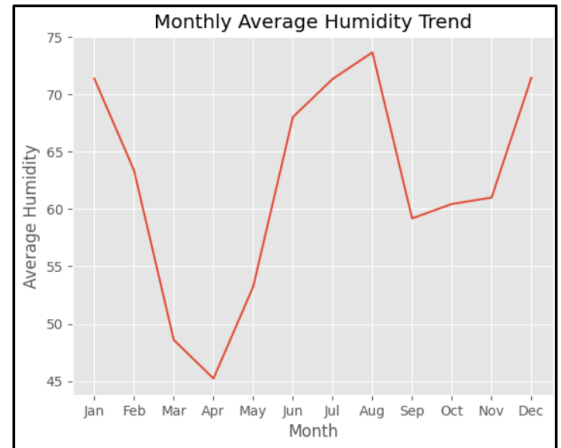


1. Overall Seasonal Pattern

- The graph shows a distinct seasonal variation in humidity levels throughout the year.
- Humidity is relatively high at the beginning and end of the year and varies significantly during the middle months.

2. Dry Period (Late Winter–Early Summer)

- Humidity declines steadily from January to April.
- The lowest average humidity occurs in April, indicating the driest atmospheric conditions.
- This period corresponds to dry winter and pre-summer months.



3. Monsoon Influence

- From May to August, humidity rises sharply.
- Peak humidity is observed in August, which aligns with the monsoon season.
- Increased rainfall and cloud cover lead to higher moisture content in the air.

4. Post-Monsoon Transition

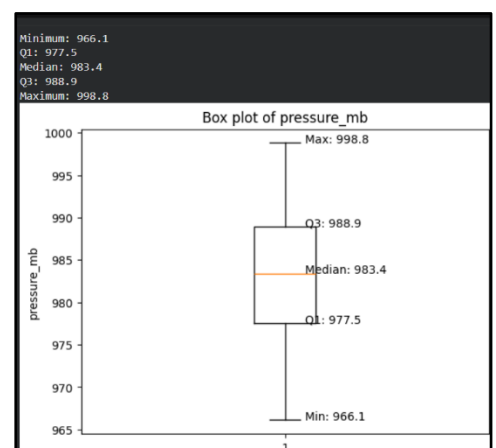
- A noticeable drop in humidity occurs in September, marking the withdrawal of the monsoon.
- Humidity then gradually increases from October to December, reflecting cooler conditions with moderate moisture.

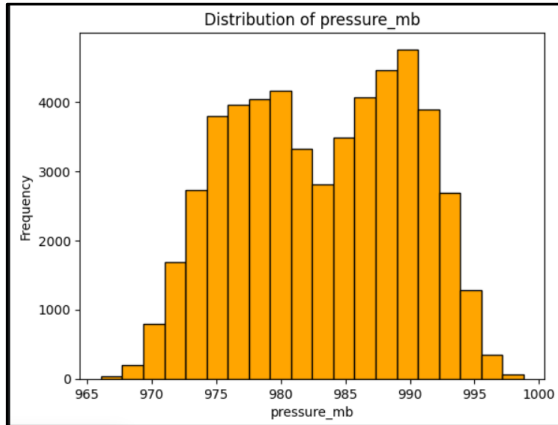
5. Climatic Insight

- The trend highlights the strong impact of seasonal weather patterns on humidity levels.
- Variations in humidity help explain changes in thermal comfort, pollution dispersion, and air quality.

Pressure

50% of the pressure values lie between 977.5 mb- 988.9 mb. This narrow range shows that most pressure readings are closely concentrated, reflecting stable atmospheric conditions.





Pressure values between 966 mb and 998 mb suggest moderate variability in weather conditions. Lower values around 966 mb are typically associated with low-pressure systems, which often bring cloudy, rainy, or stormy weather.

Higher values near 998 mb indicate relatively stable and clearer atmospheric conditions.

- Atmospheric pressure values are mainly concentrated between 970 mb and 995 mb, indicating stable and safe weather conditions.

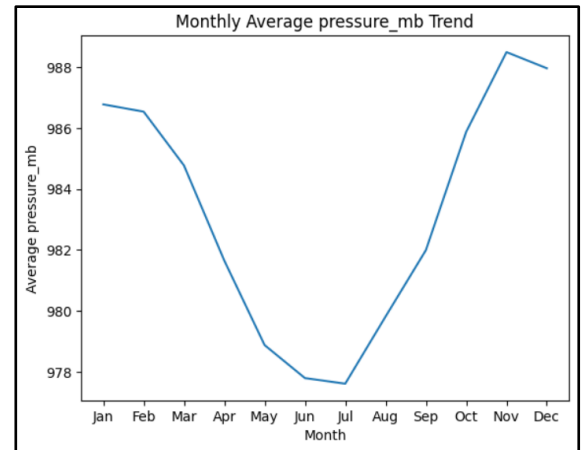
- The distribution is fairly balanced with a slight left skew.

1. Seasonal Pattern

- Pressure values gradually decrease from January to July, indicating changing seasonal conditions.
- The lowest average pressure is observed during the mid-year months, reflecting the dominance of low-pressure systems.

2. Pre-Monsoon to Monsoon Transition

- From March to July, pressure shows a consistent decline.
- This decrease corresponds to the pre-monsoon and monsoon period, when rising temperatures and increased moisture lead to lower surface pressure.
- The minimum pressure around June–July indicates peak monsoon influence.

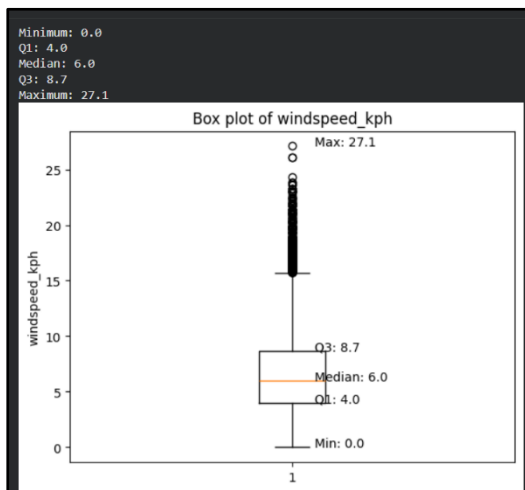


3. Post-Monsoon and Winter Onset

- From August onwards, atmospheric pressure increases steadily.
- The highest pressure values are recorded during November and December, indicating stable winter conditions.
- This rise reflects the establishment of cooler and high-pressure systems.

4. Data Stability Note

- No extreme or abnormal pressure values are observed in the dataset.
- The absence of outliers indicates reliable pressure measurements and ensures accurate seasonal trend interpretation.



Windspeed

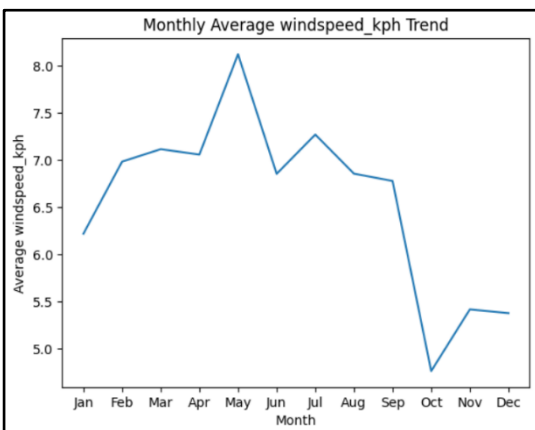
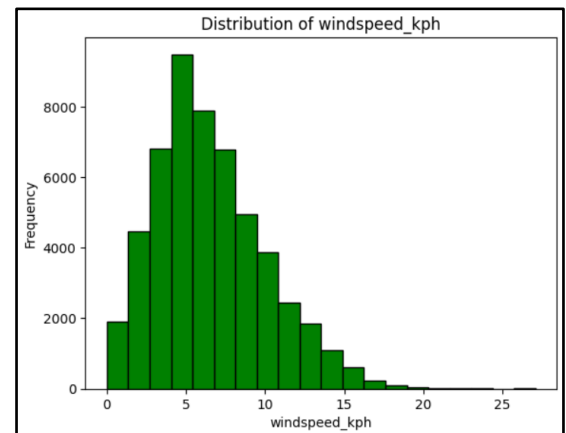
50% of windspeed values lie between 4.0 kph - 8.7 kph, indicating that typical wind speeds are moderate and consistent.

High-value outliers are present, representing occasional strong wind events, possibly due to storms or seasonal weather disturbances.

The distribution is positively skewed.

Windspeed values are mostly concentrated between 2 kph and 10 kph, indicating that light to moderate winds dominate the dataset.

The highest frequency occurs around 4–6 kph, which represents the most common wind conditions.



1. Seasonal Pattern

- Windspeed increases gradually from January to May, indicating strengthening winds during the pre-summer period.
- The highest average windspeed occurs in May,

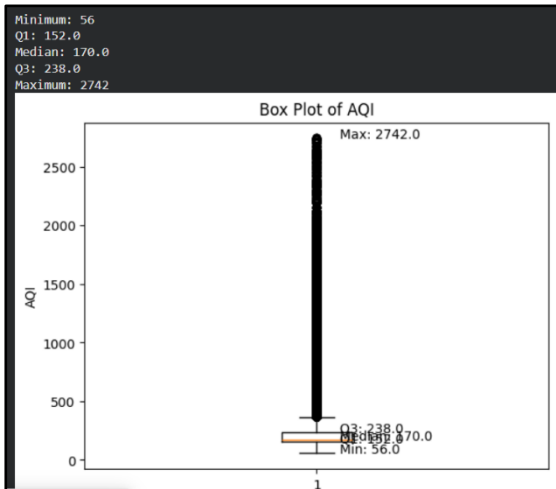
representing peak windy conditions.

2. Summer to Monsoon Transition

- After May, windspeed shows a fluctuating but slightly declining trend from June to September.
- This variation corresponds to the monsoon season, where wind patterns become irregular due to changing pressure systems and rainfall.
- A noticeable drop in September suggests calmer conditions toward the end of the monsoon period.

3. Post-Monsoon and Winter Conditions

- From October onwards, windspeed remains relatively low and stable.
- November and December show reduced wind activity, indicating calmer atmospheric conditions during winter.
- These months experience the lowest average windspeed in the dataset.



AQI

The box plot shows that AQI values range from a minimum of 56 to a maximum of 2742, indicating very high variability in air quality conditions across the dataset.

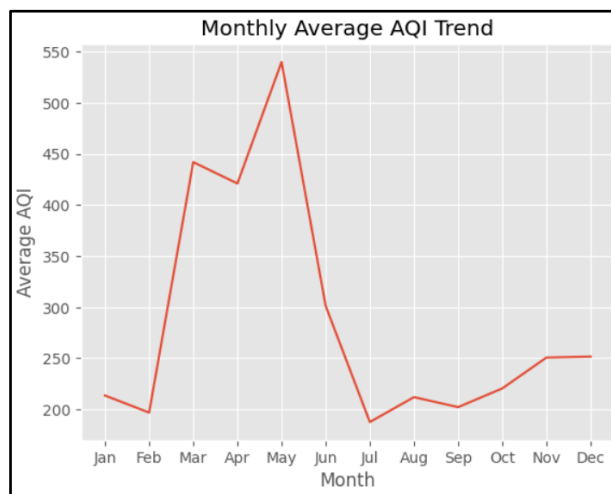
50% of AQI values lie between 152 (Q1) and 238 (Q3), showing that most observations experience elevated pollution levels.

The distribution is strongly positively skewed, with a large number of extreme high-value outliers above the upper quartile. These

outliers represent severe pollution.

1. Seasonal Pattern

- The graph shows a clear seasonal variation in air quality across the year.
- AQI values remain moderate during January and February, indicating comparatively better air quality in early winter.
- A sharp rise in AQI is observed from March to May, reflecting worsening air quality during the pre-summer period.



2. Pre-Summer Pollution Peak

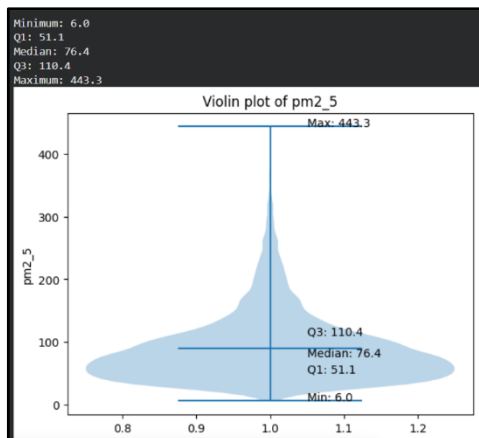
- The highest average AQI occurs in May, indicating severe pollution conditions.
- This peak may be attributed to factors such as increased dust, higher temperatures, reduced wind dispersion, and human activities.
- The elevated AQI values during this period indicate poor air quality and potential health risks.

3. Monsoon Improvement

- From June to September, AQI levels show a significant decline.
- The improvement in air quality corresponds to the monsoon season, where rainfall and stronger winds help remove pollutants from the atmosphere.
- July records the lowest average AQI, indicating the cleanest air conditions in the dataset.

4. Post-Monsoon and Winter Rise

- After September, AQI values gradually increase from October to December.
- This rise reflects the return of stable atmospheric conditions, lower wind speeds, and increased pollutant accumulation.
- Late-year months show moderate pollution levels, though lower than the pre-summer peak.



PM2.5

PM2.5 concentrations range from a minimum of 6.0 $\mu\text{g}/\text{m}^3$ to a maximum of 443.3 $\mu\text{g}/\text{m}^3$, indicating very high variability in particulate pollution levels.

Middle 50% of values lie between 51.1 $\mu\text{g}/\text{m}^3$ - 110.4 $\mu\text{g}/\text{m}^3$, indicating that most observations fall within the moderate to unhealthy pollution range.

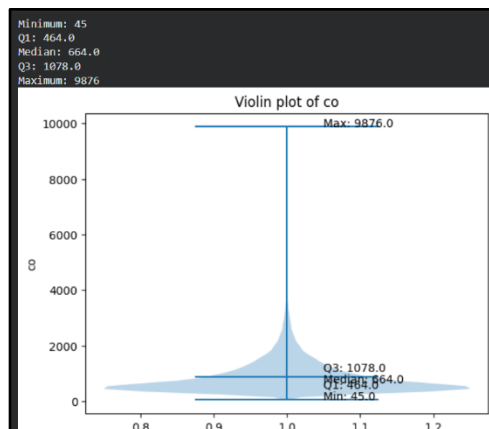
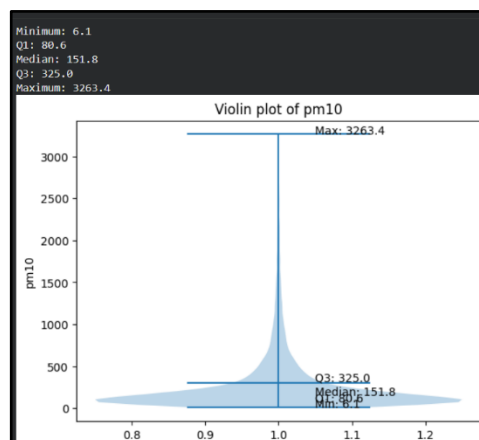
The distribution is strongly right-skewed. This reflects the presence of severe pollution episodes.

PM10

The violin plot shows a highly right-skewed distribution of PM10 concentrations, indicating the presence of extreme pollution episodes.

The wide interquartile range (80.6- 325.0), reflecting substantial variability in PM10 concentrations.

The dense width of the violin around lower PM10 values indicates that the majority of readings cluster below the upper quartile.



CO

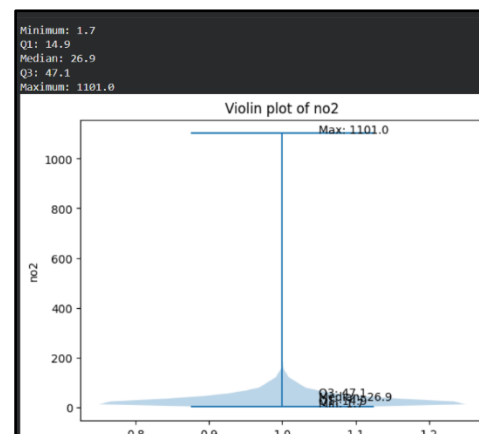
The violin plot indicates a strong right-skewed distribution.

The interquartile range (Q1 = 464.0 to Q3 = 1078.0) is wide, reflecting substantial variability in CO levels.

The violin plot reveals that CO levels are generally moderate but punctuated by rare, extremely high pollution events.

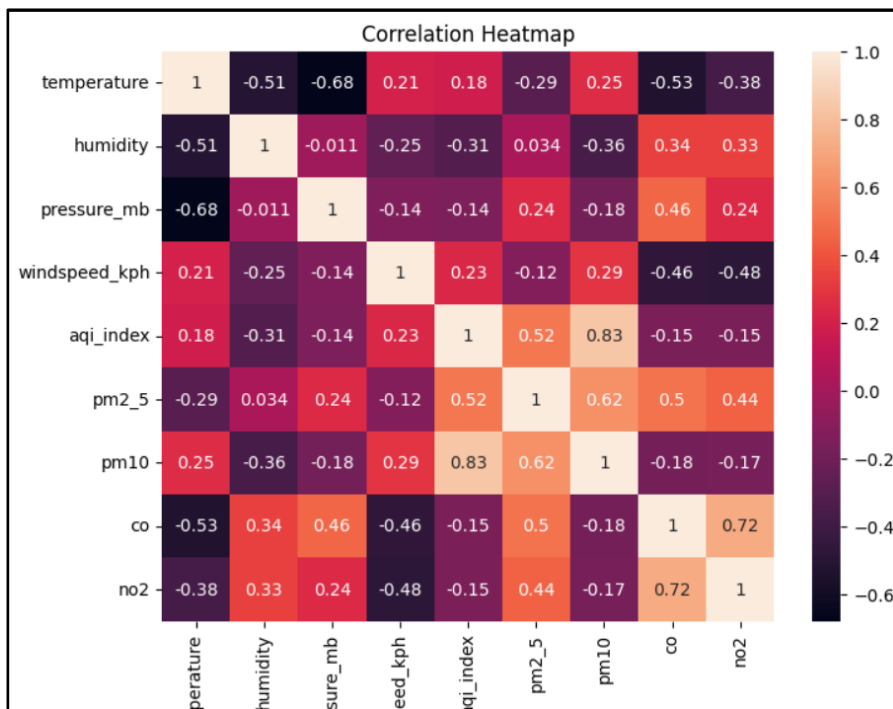
NO2

The violin plot shows a positively skewed (right-skewed) distribution.



The interquartile range (Q1 = 14.9 to Q3 = 47.1) is moderately wide, reflecting noticeable variability in NO₂ levels. The violin plot highlights that NO₂ concentrations are generally low to moderate but are significantly impacted by episodic high-pollution events.

Correlation



Relationship between Weather Variables

- Temperature and pressure has a strong negative correlation (-0.68), indicating that higher temperatures are generally associated with lower atmospheric pressure.
- Temperature and humidity are moderately negatively correlated (-0.51), suggesting that warmer conditions tend to coincide with lower relative humidity.
- Wind speed shows a weak positive correlation with temperature (0.21), indicating that warmer conditions are often associated with slightly higher wind speeds. This occurs because increased surface heating enhances atmospheric convection, which promotes air movement and vertical mixing; and a weak negative relationship with humidity (-0.25), suggesting that stronger winds help disperse moist air and reduce moisture accumulation near the surface.

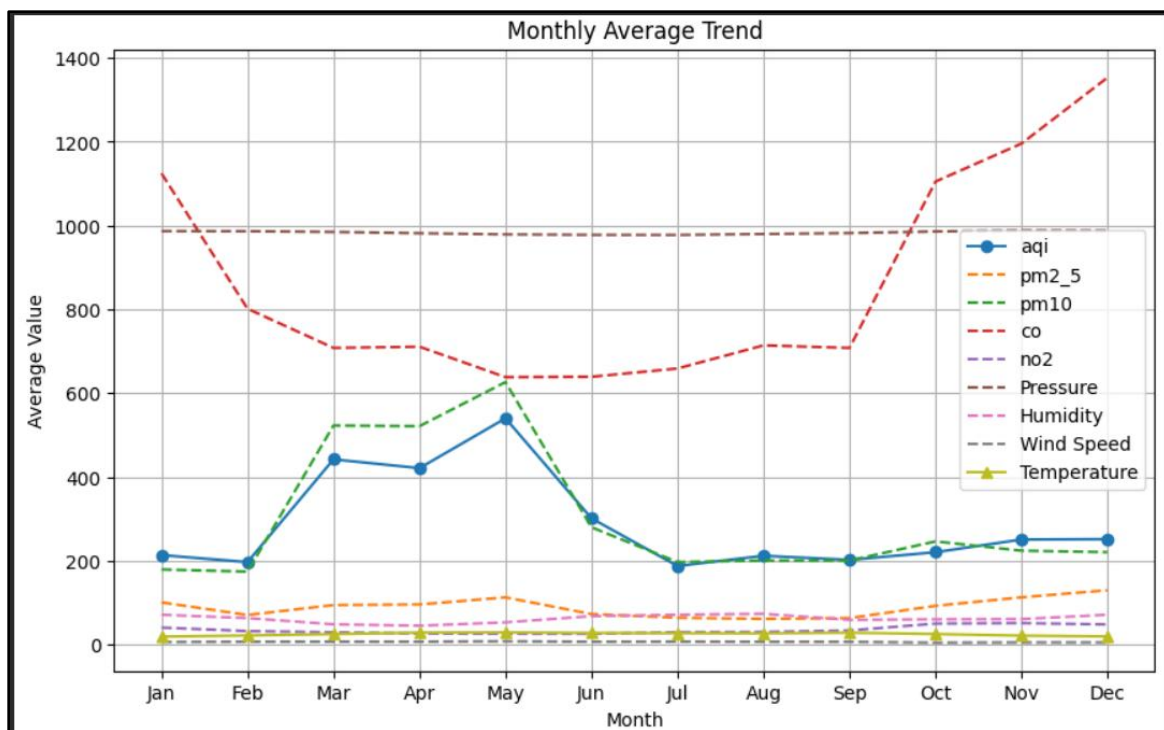
Relationship between Pollutants and AQI

- Particulate matter (PM10 and PM2.5) shows a strong positive relationship with AQI, indicating that an increase in particulate concentration directly leads to deterioration in air quality. The high

correlation of AQI with PM10 (0.83) and PM2.5 (0.52) confirms that particulate pollutants are the primary drivers of AQI levels.

- PM10 has a stronger influence on AQI than PM2.5 in the dataset, suggesting that coarse particles from sources such as road dust, construction activity, and resuspended soil significantly impact overall air quality.
- Gaseous pollutants such as CO and NO₂ show weak correlations with AQI, indicating that while they contribute to pollution, their direct impact on AQI is less pronounced compared to particulate matter.

Monthly analysis



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temp.	Low	Low-moderate	Mod	High	Very High	High	High	Mod	High	Mod	Mod	Low
Wind	Mod	High	High	High	Very high	High	High	High	Mod	Low	Low	Low
Humidity	Very high	High	Mod	Low	Mod	High	Very high	Very high	Mod	Mod	Mod	High
Pressure	Very high	Very high	High	High	Mod	Low	Low	Mod	Mod	High	Very high	Very high
AQI	Poor	Poor	Hazard	Hazard	Hazard	Very poor	Poor	Poor	Poor	Very poor	Very poor	Very poor
PM2.5	Very high	High	Mod	Mod	Mod	Low	Low	Low	Low	Mod	High	Very high
PM10	Very high	High	Mod	Mod	Mod	Low	Low	Low	Low	Mod	High	Very high
CO	High	High	Mod	Mod	Mod	Low	Low	Low	Low	Mod	High	High
NO2	Very high	High	High	Mod	Mod	Mod	Mod	Mod	Mod	High	Very high	Very high

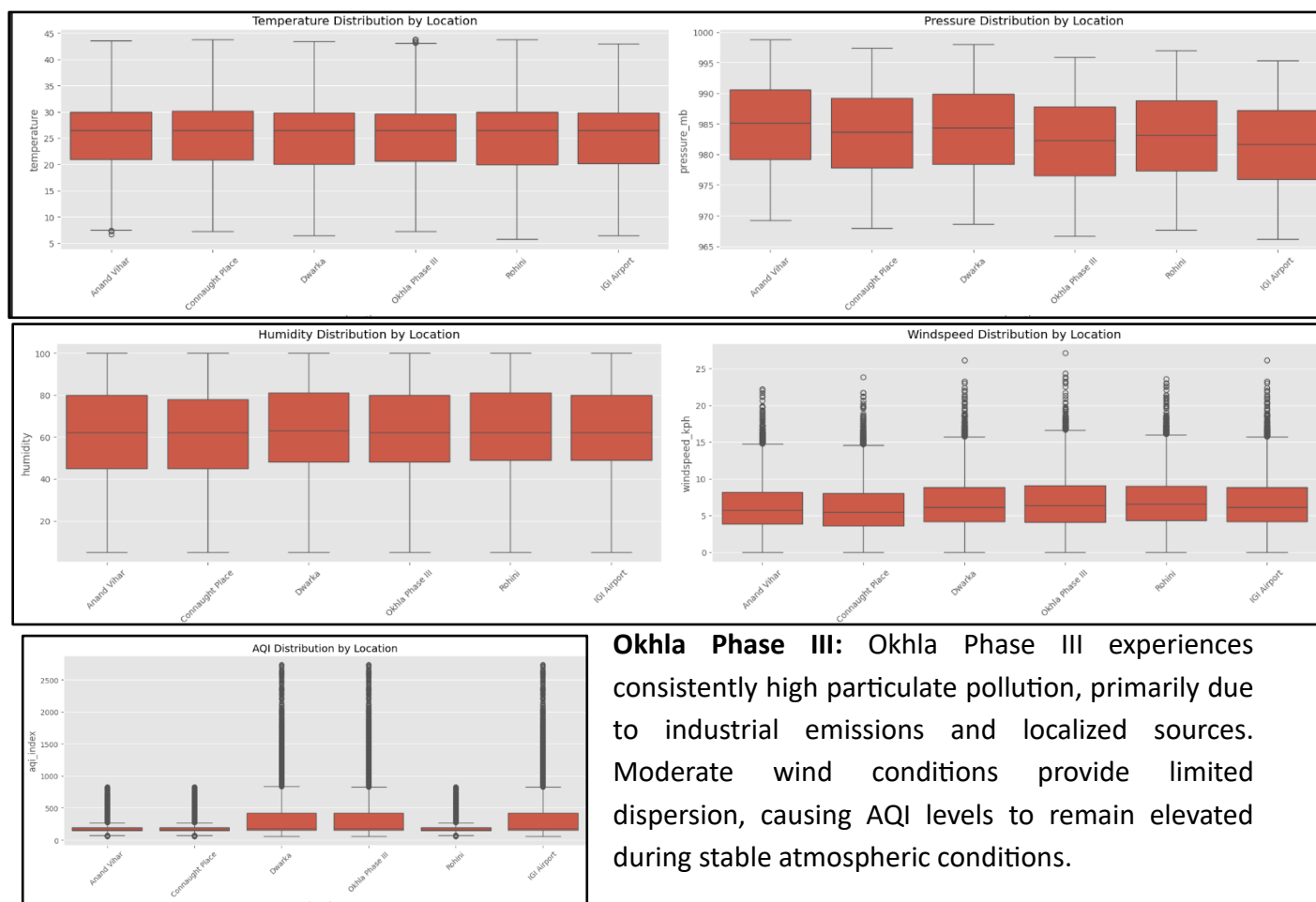
January: Cold, stable conditions; high pollution
February: Slight warming; pollution remains elevated
March: Transitional weather; pollution increases
April: Hot, dry conditions; dust-related pollution
May: Extreme heat; persistently high pollution
June: Monsoon onset; pollution levels decline
July: Strong monsoon influence; lowest pollution
August: Continued rainfall; reduced pollution
September: Post-monsoon transition; moderate pollution
October: Calm conditions; pollution buildup begins
November: Stable atmosphere; severe pollution
December: Cold, stagnant air; extreme pollution

Location based analysis

Anand Vihar: Anand Vihar records the highest pollution levels among all stations, with persistently poor to very poor AQI and elevated PM_{2.5}, PM₁₀, and NO₂ concentrations. Low wind speeds and high atmospheric pressure, combined with heavy traffic and interstate transport activity, restrict pollutant dispersion and result in severe pollution accumulation.

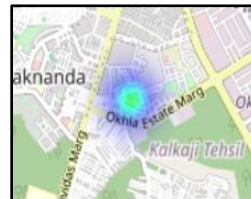
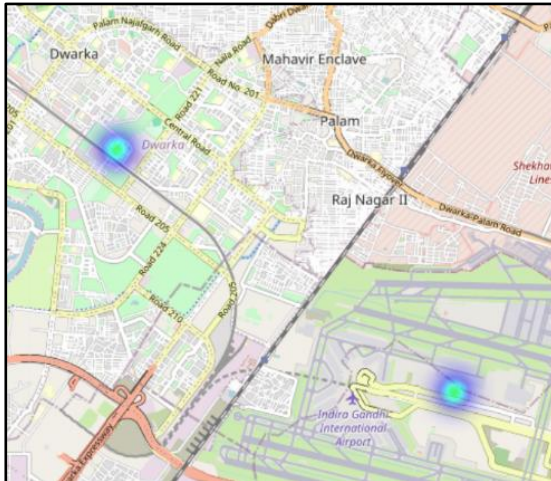
Connaught Place: Connaught Place exhibits moderate to high AQI levels, influenced mainly by vehicular emissions and commercial activity. Although wind speeds are slightly better than eastern Delhi, periods of high pressure and reduced ventilation lead to noticeable pollutant buildup, particularly during winter and post-monsoon months.

Dwarka: Dwarka shows comparatively better air quality than central and eastern locations. Higher wind speeds, relatively open surroundings, and lower congestion contribute to improved dispersion, resulting in lower AQI and particulate concentrations, especially during monsoon months.



Rohini: Rohini displays moderate air quality, with pollution levels strongly influenced by seasonal weather variations. AQI improves during periods of higher wind speed and lower pressure but deteriorates under calm winter conditions.

IGI Airport: The IGI Airport station records moderate AQI levels, with pollution influenced by fuel combustion and vehicular movement. Relatively higher wind speeds help prevent extreme pollutant accumulation, leading to better air quality compared to densely populated urban stations.



Anand Vihar: Severe pollution; poor dispersion conditions

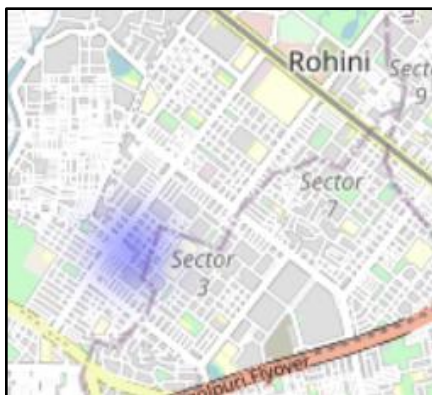
Connaught Place: Traffic-driven moderate to high pollution



Dwarka: Better dispersion; comparatively cleaner air

Okhla Phase III: Industrial influence; high particulate pollution

Rohini: Moderate pollution; strong seasonal variation



IGI Airport: Combustion-related pollution; moderate AQI

RECOMMENDATIONS

- Strengthen pollution control measures at Anand Vihar and Okhla Phase III, focusing on traffic regulation and industrial emission monitoring.
- Implement stricter vehicular emission checks and congestion management in Connaught Place.

- Promote dust-control measures, including regular road sweeping and construction-site regulation, across all stations.
- Increase green buffers and roadside plantations in high-pollution areas to reduce particulate concentration.
- Use weather-based early warning systems to manage pollution during high-pressure and low-wind conditions.
- Encourage remote working and public transport usage during severe pollution episodes, especially in winter.
- Enhance continuous air quality monitoring at Rohini, Dwarka, and IGI Airport to track seasonal variations.
- Strengthen public awareness regarding health risks during high AQI periods.

CONCLUSION

- Weather parameters significantly influence air pollution levels in Delhi.
- Particulate matter (PM_{2.5} and PM₁₀) is the primary contributor to AQI deterioration.
- High temperature and dry conditions during pre-summer months lead to severe pollution.
- Monsoon conditions improve air quality through rainfall and enhanced dispersion.
- Winter months experience severe pollution due to low wind speed and high pressure.
- Locations with high traffic and industrial activity show consistently poor air quality.
- Open and well-ventilated locations exhibit comparatively lower pollution levels.

- Strong correlations confirm the role of meteorological factors in pollution behavior.